

Real Estate Automation

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Abstract

The research objective aims to take a step towards addressing the fragmentation of the tech stack in the real estate industry. After preliminary literature and interviews, I noticed the absence of industry standardised tech stack for the targeted group, and how the novel tools don't work for an industry standard, but rather as niche solutions for narrower targets.

The impact of the work will provide a solution for the target group that covers multiple aspects of daily repetitive work. The objective is to design and implement an automated system that centralizes incoming property information, structures it into a standardized database format, and applies a predictive valuation model to rank and surface the most promising real-estate investment opportunities.

Investment funds, real estate agencies, individual investors, family offices, wealth/asset managers, are the groups that could benefit from the solution proposed. These players suffer from isolation and spread in their tech stack for daily use. The centralization of these will consolidate the use and provide a system capable of streamlining processes.

Literature Review

Real Estate Market Context and Investment Dynamics

“The number of new housing units created per household has been declining since 2018, reaching its lowest level since 2014 in 2023, which has added pressure to housing prices” Page 11, BBVA Research. *Source 5, Page 11, BBVA Research.*

New units have been declining, making opportunities more valuable. Therefore it's key for players such as funds or investors to move fast in a scarce real estate environment.

“In 2024, real estate investment in Spain reached approximately €14 billion, with Madrid accounting for 31% of the total volume.” *Source 5, Page 15, CBRE*

“Investment funds and private investors, including family offices, comprised nearly a third of the total investment volume in 2024[...]”. *Source 5, Page 15, CBRE*

“Residential assets accounted for over 30% of the total investment volume, amounting to €4.3 billion”. *Source 5, Page 15, CBRE*

Digital Transformation and PropTech in Real Estate

“PropTech companies, spanning a wide range of technological applications, have introduced significant efficiencies across the property lifecycle.” *Source 5, Page 4.*

“In response to these changes, Spanish Real Estate companies are increasingly leveraging advanced technologies, including AI, smart data, and marketplaces, to identify opportunities and streamline various aspects of the industry such as design, development, and brokerage.” *Source 5, Page 4.*

Europe leads global PropTech (50% of companies worldwide). Focus on the residential sector, countries with high demands and elevated prices per square meter such as Spain. *Source 5, Page 23, 2025 European prop tech report.*

“Business model distribution. 60% b2b + b2b2c. 40% b2c.” *Source 5, Page 24.*

B2B Analysis

28 valuation companies. 135 internal operations (33%). *Source 5, Page 25*

Tech Adoption, Tools, and Operational Challenges

Adoption. 74% report having increased their use of tech.

Commonly used tech: Valuation & reporting 64% ; automation 28% ; platforms 24%. *Source 5, Page 29, Maps PropTech API.*

Tech Stack. Solutions used. Most professionals rely on a small set of core tools

61% use valuation tools. Adoption is still centered around basic and operational tools. Most agencies have 3-5 solutions. Rooms for growth in complementary services and automation tools. *Source 5, Page 30, Maps PropTech API-*

Where’s the need?. “In which processes do you need help?” 47% Automation top priority. Doubling from last year. Repetition is a major burden. *Source 5, Page 31, Maps PropTech API.*

“62% find it difficult to identify the right proptech tool, Spain is a fragmented market with many options. Choosing the right tool requires a deep understanding of internal processes and an alignment with team, costs and workflows”. **Source 5, Page 32, Maps PropTech API.**

Key for validation of the tool proposed. Centralization of solutions + understanding of internal processes (validated by experience)

20% feel that the tools don't match their needs. “Too many tools, not enough clarity”. **Source 5, Page 32, Maps PropTech API.**

The real estate industry is increasingly adopting specialized AI and Automated Valuation Models (AVMs): Established AVMs: These use Hedonic Pricing Models (HPM), Machine Learning (ML), Artificial Neural Networks (ANN), and Geospatial Models (GIS) to estimate market values. **Source 1, Automatic valuation models.**

Automation, AI, and Decision Support in Real Estate

Asset identification and opportunity profiling. **Source 5, Page 19.**

Technologies that funds would like to increase investments in 2025. AI: 56% plan to allocate more funds in 2025 compared to prior year. We assume a trend continues. **Source 5, Page 22, Zucua Ventures.**

Main uses of AI in Real Estate. To know / find ; to generate ; to conversate

1. Identification and profiling: generative search models, opportunity identification, opportunity qualification.
2. Operational efficiency: task and process optimization.
3. Decision making: Dashboards, risk assessment, analysis of options, scenario modeling.

Source 5, Page 47, Snapland.

Commercial Competitors

1. IntelICRE

AI-powered underwriting and instant property data aggregation. It differentiates itself through marketing automation, generating brochures, property websites, and flipbooks autonomously. At a cost of roughly \$69 per month, it targets mid-market brokers and investors who need to evaluate deals in seconds.

2. HouseCanary / CanaryAI

Focuses on high-accuracy valuation and forecasting. Its AVMs achieve sub-3% error rates, significantly better than the industry standard. Its CanaryAI assistant allows users to query 136 million property records using plain English.

3. Cherre

Focuses on data normalization and infrastructure. It uses a "Universal Data Model" to connect over 3.3 billion addresses into a single knowledge graph, helping large institutions and REITs eliminate fragmented internal databases.

4. ATTOM AVM

Focuses on a multi-model approach that selects the optimal valuation algorithm based on a property's specific geographic area. It targets mid-market firms by offering a balance between free consumer tools and expensive lender-grade AVMs.

5. CoreLogic

Focuses on a single-model AVM using advanced network graph technology for extreme consistency across 99.9% of U.S. properties. It uniquely integrates natural disaster risk analysis (hurricanes, wildfires, floods) into its valuations.

6. Quantarium QVM

Focuses on a self-learning AI engine that processes 900 factors. It is notable for its TeraLook visual technology, which incorporates image analysis of property and neighborhood conditions into its values.

92% of firms are piloting AI but only 5% have achieved their goals, largely due to legacy infrastructure and fragmented ingestion. Most existing solutions focus on the "analysis" stage but ignore the "ingestion" stage, specifically the messy data coming from unstructured sources.

Source 6, v7 Labs – AI in Real Estate: Key Use Cases, Solutions, and Challenges, Jan 2026.

Limitations of Existing Approaches and Research Gap

Cost perception of tools. “A small minority consider proptechs affordable. This highlights a sensitive issue around pricing and perceived ROI.”

“Value-for-money is essential, tools that are simple, integrated, and impactful will justify higher investment.”

It’s more about perceived value and ease of use than cost. 41% of Professionals consider it acceptable to spend between 150 and 400 euros a month on tech. *Source 5, Page 33, Maps PropTech API.*

Barriers to adoption. Cost is still the major one (60%), followed by lack of information (45%).

Impact of tech. 80% of respondents said technology improved their processes. Tech continues to deliver clear value in internal operations, where impact is most tangible.

Looking Ahead. Adoption already happened now the focus is in making these tools work. Optimizing usage, creating real value, embedding into daily operations. The sector seems to be entering a consolidation phase, where decisions are more strategic than impulsive.

Source 5, Page 47, Snapland.

System Architecture & Tech Stack

Backend

The backend is divided into two services: Node.js and FastAPI.

Node.js is the main application layer. It is used because it handles asynchronous operations efficiently and works well as a central coordination layer.

It receives requests from the frontend, manages authentication and routing, and acts as the API gateway. When a model is requested, it sends structured property data to the FastAPI service and returns the results to the client.

FastAPI is used for the machine learning service because it runs in Python, which is the language used for the data science stack. It loads the trained valuation model, receives property features and generates price predictions. The trained model is saved using joblib and loaded by the service when needed. FastAPI exposes endpoints that Node.js calls through HTTP requests.

Finally Uvicorn runs the FastAPI service. It is the server that handles incoming HTTP requests and allows the ML service to respond efficiently and concurrently.

Database

PostgreSQL is selected as the primary database system due to its robustness, reliability, and suitability for structured real estate data. The project requires storing relational information such as deal information, its sources and the predictions that will be performed with the valuation model. Which naturally fits a relational schema with strong ACID compliance. PostgreSQL ensures data integrity through constraints, indexing, and transactional consistency.

Additionally, its compatibility with Python through psycopg2 and SQLAlchemy enables seamless integration with the machine learning pipeline. Furthermore, PostgreSQL can be extended with spatial capabilities (PostGIS), which is particularly relevant for geolocation-based real estate analysis.

Overall, provides scalability, analytical flexibility, and production-level reliability appropriate for a data-driven real estate investment platform.

Preliminary Tables:

MESSAGES		
uuid	id	PK
string	channel	
string	source_id	
string	sender	
datetime	received_at	
text	raw_subject	
text	raw_body_text	
text	raw_body_html	
jsonb	attachment_urls	
datetime	created_at	

DEALS		
uuid	id	PK
uuid	message_id	FK
text	address	
text	city	
text	country	
text	property_type	
float	size_sqm	
integer	bedrooms	
integer	bathrooms	
integer	floor	
float	asking_price	
string	currency	
text	url	
text	broker_name	
text	broker_contact	
datetime	created_at	

PREDICTIONS		
uuid	id	PK
uuid	deal_id	FK
float	predicted_price	
string	model_version	
datetime	created_at	

SQLAlchemy is used to interact with PostgreSQL using Python instead of writing raw SQL.

Database tables are defined as Python classes. SQLAlchemy translates Python operations into SQL queries. This keeps the data layer consistent and easier to maintain.

Alembic manages changes to the database structure over time. When the schema changes, migration scripts are generated so the database can be updated in a controlled way without losing data.

Data collection

Collection of the data will happen at different levels. We will collect incoming deals from various sources such as whatsapp, gmail, or property portals.

For **incoming deals from messaging platforms** (whatsapp / gmail) we will first store the data in our 'Messages' table (specific variables can be seen in the image below). Which will then be transformed into deals and stored in their respective table.

For **deals directly scraped from a property portal**, either using its API or web scraping, the data will be stored in our 'Deals' table (specific variables stored can be seen in the image below).

Data Processing and Machine Learning

Data processing is done in Python using **pandas** and **numpy** for cleaning, transformation, and numerical operations.

Scikit-learn is used to train machine learning models that predict property prices based on its characteristics. After training, the model is serialized using **joblib** and deployed inside the **FastAPI** service. When new property features are sent to the service, it generates a predicted value in real time.

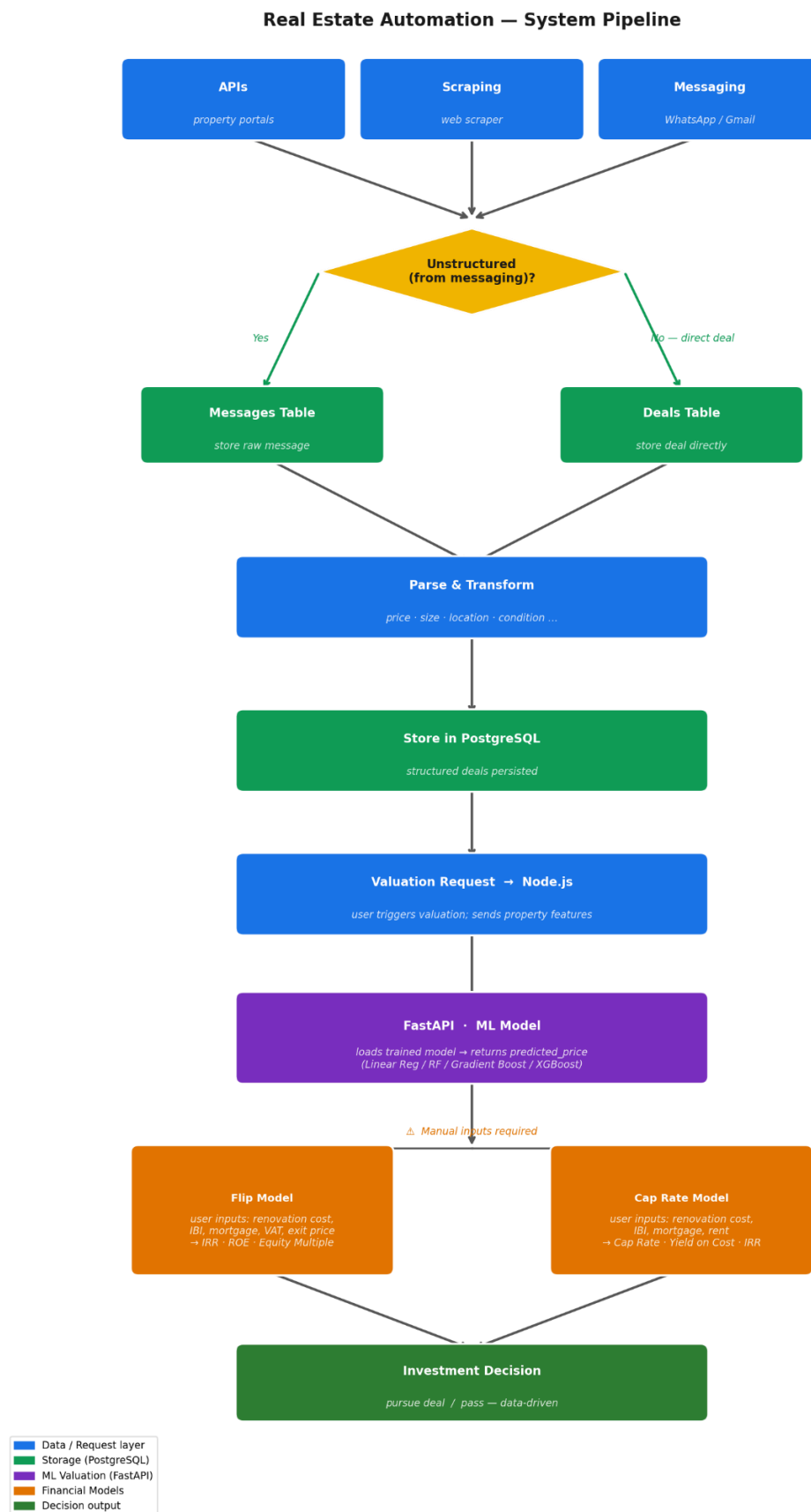
Pipeline

The system follows a structured flow.

1. First, data enters through APIs, scraping, or messaging platforms.
2. If coming from messages it goes to the respective table, and if it is directly a deal coming from a portal or API it goes to the 'Deals' table.
3. Both are parsed and transformed into structured variables such as price, size, location, and condition among others and deals are stored in PostgreSQL.
4. When a valuation is requested, Node.js sends property features to FastAPI. FastAPI loads the trained model and returns a predicted price.
5. Regarding the Flip and Cap Rate models. They require the user to plug information manually as inputs for the model to generate outputs.

In simple terms, the system collects property data, structures it, stores it, predicts its value, and evaluates whether it is a good investment.

Flow Diagram



Analysis & Algorithms

Search Optimization

Collected data is already extracted and structured. Ready for quantitative analysis, contact the owner to streamline the acquisition process and presentation level.

Valuation Model (Machine Learning).

Predict the price given the characteristics of the asset.

Target variable:

- Price

Models to be tested:

- Linear regression
- Random Forest Regressor
- Gradient Boosting
- XGBoost

Evaluation metrics:

- MAE
- RMSE
- R^2
- MAPE

Financial Models. (At present, excel based).

Flip

Second hand apartments for sale are scrapped → Apply renovation model → Compare exit prices and evaluate if worth pursuing the opportunity.

Inputs: project time (in months), renovation cost per square meter, IBI, utilities, mortgage (if there is), commissions, VAT, potential exit price (market comparison of similar apartment pre vs post renovation).

Outputs:

- IRR
- Return on Equity: $\text{Equity Profit} / \text{Equity Invested}$
- Equity Multiple: $\text{Total Cash Returned} / \text{Equity Invested}$
- Gross Margin: $\text{Profit} / \text{Total Project Cost}$

CapRate

Second hand apartments for sale are scrapped → Apply renovation model → Compare cap rate pre vs post renovation and evaluate if worth pursuing the opportunity.

Inputs: project time (in months), renovation cost per square meter, IBI, utilities, mortgage (if there is), commissions, VAT, potential new rent (market comparison of similar apartment pre vs post renovation).

Outputs:

- Cap Rate (Current and post renovation)
- Yield on cost: *Net Operating Income (post) / Total Project Cost*
- Market cap rate (zone benchmark)
- Value creation spread: *Yield on Cost – Market Cap Rate*
- IRR

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<https://www.v7labs.com/blog/best-ai-tools-for-real-estate>

Final Draft What to add/change

Literature review

- Clean up todo. Mejorar el escrito.
- La clave es encontrar los puntos que quiero dejar en evidencia.
 - Pocos jugadores tienen acceso a softwares centralizados
 - Las herramientas no abarcan el proceso completo.
 - Centralizar módulos y herramientas puede derivar en una ganancia de eficiencia como de mayor nivel de análisis.

Methodology

- Definición del proyecto completo.
- Cómo vamos a llevar a cabo el proyecto.
- Explicar como funciona todo como se conecta todo desde un nivel mas abstracto, menos tecnico

System Architecture & Tech Stack – Que tecnologías usamos en cada capa del proyecto. Como funcionan.

- Backend
- Database
- Data Collection
 - Scraping
 - API
- Data Processing & Machine Learning

Pipeline Completo – Ejemplo de cómo fluye la información dentro del sistema. De principio a fin.

Analysis & Algorithms

- Search optimization (ETL, scraping and API, structure, and load)
- Valuation Model
 - Modelos
 - Mostrar el feature engineering
 - Cleaning
 - Processing
 - Visualizaciones
 - Eval metrics
 - [4 Effective Ways to Visualize XGBoost Trees](#)
 - Conclusiones
- Financial Model
 - Flip (explain how it works)
 - Inputs and outputs
 - Conclusions
- Analytics
 - Agregar visualizaciones: Tablas base de datos (llenas)
 - Exploratory data Analysis Sobre la base de datos (dashboards)

Results & Conclusions

- How does this software deliver efficiency gains with 3 crucial modules?
- Quantify it.
- Agregar feedback de profesionales (5 entre arg y españa)
- Posibles futuros del proyecto.
 - Target B2B. Fondos de inversión, mejorar la tecnología. Crecer vertical
 - Target B2C. Inversores retail. Sin infraestructura. Como solución de donde empezar. Freemium modelo de negocio, tiers. Juntar una base de usuarios interesante y que luego inmobiliarias quieran feature en la plataforma y también quieran acceder a las data de esos usuarios.